

ON A GENERALIZED MATRIX-VARIATE HYPERGEOMETRIC DISTRIBUTION

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1. Introduction

The generalised hypergeometric distribution or a general family of statistical probability distributions was introduced by Mathai and Saxena (1966), from which almost all the classical probability distributions are obtained as special cases. A similar multivariate probability distribution is considered in this paper from which the multivariate Beta and Wishart distributions are obtained as special cases, as are certain functions of the multivariate normal matrices.

In section 2 we consider a probability density function of a symmetric random matrix X . The incomplete distribution function, the distributions of the roots of the matrix X , the distribution of the ratio $|X_1||X_2|^{-1}$ and the distributions of a random matrix with random matrix parameters are investigated.

Some of these special multivariate distributions are studied by Anderson (1958), Roy (1957), Khatri (1959) and Olkin and Rubin (1964), because of the importance and the practical application of these in testing certain statistical hypotheses.

2. A generalized matrix-variate hypergeometric distribution

Let $X: p \times p$ be a random matrix whose probability density function is,

$$(2.1) f_1(X) = \begin{cases} C|X|^{n-\frac{1}{2}(p+1)} {}_2F_1(\alpha, \beta; \theta; -AX), & X > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where ${}_2F_1(\cdot)$ is a hypergeometric function with matrix argument (Constantine (1963)) and

$$C = \frac{|A|^n \Gamma_p(\alpha) \Gamma_p(\beta) \Gamma_p(\theta-n)}{\Gamma_p(n) \Gamma_p(\theta) \Gamma_p(\alpha-n) \Gamma_p(\beta-n)}$$

for $\text{Re}(\theta-n) > \frac{1}{2}(p-1)$, $\text{Re}(\alpha-n) > \frac{1}{2}(p-1)$ and $\text{Re}(\beta-n) > \frac{1}{2}(p-1)$. The parameters α , β and θ are restricted to take values for which $f_1(X)$ is positive.

Lemma 2.1

$$\int_{X>0} f_1(X) dX = 1.$$

Proof:

Herz (1955) defined the ${}_2F_1$ function by the integral representation

$$(2.2) \quad {}_2F_1(\alpha, \beta; \beta; -Y) = \frac{\Gamma_p(\theta)}{\Gamma_p(\alpha)\Gamma_p(\theta-\alpha)} \int_0^1 |I+SY|^{-\beta} |S|^{\alpha-\frac{1}{2}(p+1)} |I-S|^{\theta-\alpha-\frac{1}{2}(p+1)} dS.$$

In (2.1) let $X = A^{-\frac{1}{2}} Y A^{-\frac{1}{2}}$ with $J(X \rightarrow Y) = |A|^{-\frac{1}{2}(p+1)}$ (Deemer and Olkin (1951)) then

$$(2.3) \quad \int_{X>0} f_1(X) dX = C |A|^{-n} \int_{Y>0} |Y|^{n-\frac{1}{2}(p+1)} {}_2F_1(\alpha, \beta; \beta; -Y) dY.$$

Using (2.2) and changing the order of integration, we obtain from (2.3)

$$(2.4) \quad C |A|^{-n} \frac{\Gamma_p(\theta)}{\Gamma_p(\alpha)\Gamma_p(\theta-\alpha)} \int_0^1 \int_{Y>0} |Y|^{n-\frac{1}{2}(p+1)} \cdot |I+SY|^{-\beta} |S|^{\alpha-\frac{1}{2}(p+1)} |I-S|^{\theta-\alpha-\frac{1}{2}(p+1)} dS.$$

Make the transformation $Y = S^{-\frac{1}{2}} M S^{-\frac{1}{2}}$ in (2.4) where the Jacobian of the transformation is $|S|^{-\frac{1}{2}(p+1)}$, then

$$C |A|^{-n} \frac{\Gamma_p(\theta)}{\Gamma_p(\alpha)\Gamma_p(\theta-\alpha)} \int_0^1 \int_{M>0} |M|^{n-\frac{1}{2}(p+1)} |I+M|^{-\beta} dM \cdot \int_0^1 |S|^{\alpha-n-\frac{1}{2}(p+1)} |I-S|^{\theta-\alpha-\frac{1}{2}(p+1)} dS.$$

Using the Beta type I and type II integrals (Olkin and Rubin (1964)) the lemma follows.

In (2.1) replace $A: p \times p$ by $\alpha^{-1} A$ and take the limit when α tends to infinity, then using the formula (James (1964))

$$\lim_{\alpha \rightarrow \infty} {}_2F_1(\alpha, \beta; \beta; -\alpha^{-1} AX) = {}_1F_1(\beta; \beta; -AX)$$

and the fact that $\lim_{\alpha \rightarrow \infty} |\alpha^{-1} A|^n \frac{\Gamma_p(\alpha)}{\Gamma_p(\alpha-n)} = |A|^n$ we obtain

$$(2.5) \quad f_2(X) = \begin{cases} \frac{|A|^n \Gamma_p(\beta) \Gamma_p(\theta-n)}{\Gamma_p(n) \Gamma_p(\theta) \Gamma_p(\beta-n)} |X|^{n-\frac{1}{2}(p+1)} {}_1F_1(\beta; \beta; -AX), & X > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Similarly by replacing $A:pxp$ by $\beta^{-1}A$ in (2.5) and taking the limit when β tends to infinity, we obtain

$$(2.6) f_3(X) = \begin{cases} \frac{|A|^n \Gamma_p(\theta - n)}{\Gamma_p(n) \Gamma_p(\theta)} |X|^{n-\frac{1}{2}(p+1)} {}_0F_1(\theta; -AX), & X > 0 \\ 0 & \text{elsewhere} \end{cases}$$

If however we replace $A:pxp$ by θA in (2.5) and take the limit when θ tends to infinity we obtain

$$(2.7) f_4(X) = \begin{cases} \frac{|A|^n \Gamma_p(\beta)}{\Gamma_p(n) \Gamma_p(\beta - n)} |X|^{n-\frac{1}{2}(p+1)} {}_1F_0(\beta; -AX), & X > 0 \\ 0 & \text{elsewhere} \end{cases}$$

In (2.7) replace A by $\beta^{-1}A$ and let β tends to infinity and we obtain

$$(2.8) f_5(X) = \begin{cases} \frac{|A|^n}{\Gamma_p(n)} |X|^{n-\frac{1}{2}(p+1)} {}_0F_0(-AX), & X > 0 \\ 0 & \text{elsewhere} \end{cases}$$

The functions $f_i(X)$, $i=2,3,4,5$ satisfy the condition $\int_{X>0} f_i(X) dX = 1$.

Let $V:pxm$ be a random matrix with the probability density function

$$(2.9) f(V) = \begin{cases} \frac{\pi^{\frac{1}{2}mp}}{\pi^{\frac{1}{2}mp}} |VV'|^{n-\frac{1}{2}m} {}_2F_1(\alpha, \beta; -AVV'), & V > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Lemma 2.2:

$$\int_{V:pxm} f(V) dV = 1.$$

Proof:

Anderson (1958) proves that if the density of $V:pxm$ is $g(VV')$, then the density of $X = VV'$ is

$$\frac{\pi^{\frac{1}{2}mp}}{\Gamma_p(\frac{1}{2}m)} |X|^{\frac{1}{2}(m-p-1)} g(X).$$

Hence the result by Lemma 2.1

The following special cases are obtained:

(i) *The Wishart distribution:*

In (2.5) put $\beta = \theta$, $n = \frac{1}{2}m$ and $A = \frac{1}{2}\Sigma^{-1}$ or in (2.8) put $A = \frac{1}{2}\Sigma^{-1}$.

(ii) *The multivariate Beta distribution:*

In (2.1) put $\beta = \theta$ and $A = I_p$ or in (2.7) put $A = I_p$.

(iii) *The multivariate folded normal distribution:*

In (2.9) put $\beta = \theta$, $A = \frac{1}{2}\Sigma^{-1}$, $n = \frac{1}{2}m$, replace $A: p \times p$ by $\alpha^{-1}A$, and take the limit when α tends to infinity.

(iv) *The general family of univariate distributions:*

In (2.1) put $p = 1$. Thus for $p = 1$ we find that almost all classical statistical univariate distributions are contained in these multivariate distributions, see e.g. Mathai and Saxena (1966) with $d = 1$.

3. The incomplete distribution function

We give a zonal polynomial expansion for the probabilities $P(S < X)$ where the matrix S has the distribution (2.1).

Theorem 3.1

$$\int_0^X |S|^{n-\frac{1}{2}(p+1)} {}_2F_1(\alpha, \beta; \theta; AS) dS = \frac{\Gamma_p(n) \Gamma_p[\frac{1}{2}(p+1)]}{\Gamma_p[n+\frac{1}{2}(p+1)]} |X|^n \cdot {}_3F_2[\alpha, \beta, n, n+\frac{1}{2}(p+1), \theta; AX] .$$

Proof:

For $R: p \times p$ positive definite it follows that (Constantine (1963)):

$$(3.1) \int_0^1 |S|^{t-\frac{1}{2}(p+1)} |I-S|^{u-\frac{1}{2}(p+1)} C_K(RS) dS = \frac{\Gamma_p(t, K) \Gamma_p(u)}{\Gamma_p(t+u, K)} C_K(R) .$$

Using (3.1) it follows that

$$(3.2) \int_0^1 |Y|^{n-\frac{1}{2}(p+1)} {}_2F_1(\alpha, \beta; \theta; BY) dY$$

$$\begin{aligned}
&= \int_0^1 |Y|^{n-\frac{1}{2}(p+1)} |I-Y|^{\frac{1}{2}(p+1)-\frac{1}{2}(p+1)} {}_2F_1(\alpha, \beta; \theta; BY) dY \\
&= \frac{\Gamma_p(n) \Gamma_p[\frac{1}{2}(p+1)]}{\Gamma_p[n+\frac{1}{2}(p+1)]} {}_3F_2[\alpha, \beta, n; n+\frac{1}{2}(p+1), \theta; B].
\end{aligned}$$

In (3.2) let $Y = P^{-\frac{1}{2}} S P^{-\frac{1}{2}}$ where the Jacobian of the transformation is $|P|^{-\frac{1}{2}(p+1)}$, then we have

$$\begin{aligned}
&\int_0^P |S|^{n-\frac{1}{2}(p+1)} {}_2F_1(\alpha, \beta; \theta; B P^{-\frac{1}{2}} S P^{-\frac{1}{2}}) dS \\
&= \frac{\Gamma_p(n) \Gamma_p[\frac{1}{2}(p+1)]}{\Gamma_p[n+\frac{1}{2}(p+1)]} |P|^n {}_3F_2[\alpha, \beta, n; n+\frac{1}{2}(p+1), \theta; B].
\end{aligned}$$

Putting $P = X$ and $B = P^{\frac{1}{2}} A P^{\frac{1}{2}}$ the result follows.

When S has the distribution (2.1), then

$$P(S < X) = C \frac{\Gamma_p(n) \Gamma_p[\frac{1}{2}(p+1)]}{\Gamma_p[n+\frac{1}{2}(p+1)]} |X|^n {}_3F_2[\alpha, \beta, n; n+\frac{1}{2}(p+1), \theta; -AX]$$

where for $p \geq 2$, $P(S < X) \neq 1 - P(S > X)$ (Constantine (1963)).

4. The distribution of the roots of a matrix

The joint distributions of the ordered characteristic roots of certain matrices have been determined by Anderson (1958). If $X: p \times p$ has a density function (2.1) the joint density of the characteristic roots of X can easily be found as

$$\begin{aligned}
(4.1) \quad g(\lambda_1, \dots, \lambda_p) &= C \frac{\pi^{\frac{1}{2}p^2}}{\Gamma_p(\frac{1}{2}p)} \prod_{i=1}^p \lambda_i^{n-\frac{1}{2}(p+1)} \\
&\quad \cdot {}_2F_1(\alpha, \beta; \theta; -A, D_\lambda) \prod_{i < j}^p (\lambda_i - \lambda_j), \lambda_1 > \dots > \lambda_p > 0
\end{aligned}$$

where $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$. The hypergeometric function ${}_2F_1(\cdot)$ with two matrix arguments is defined by James (1964).

Theorem 4.1.

If $\lambda_1 > \dots > \lambda_p > 0$ are the roots of $|X - \lambda I| = 0$, then the distribution of the largest root of X is

$$f(\lambda_I) = C \frac{\Gamma_p(n) \Gamma_p[\frac{1}{2}(p+1)]}{\Gamma_p[n+\frac{1}{2}(p+1)]} \sum_{k=0}^{\infty} \sum_K \lambda_I^{np+k-1} \cdot (np+k) \cdot \frac{(\alpha)_K (\beta)_K (n)_K}{(\theta)_K [n+\frac{1}{2}(p+1)]_K} \frac{C_K(-A)}{k!}, \quad 0 < \lambda_I < 1.$$

Proof:

In (4.1) put $\ell_i = \frac{\lambda_i}{\lambda_1}$, $i=2, \dots, p$, i.e. $D_\lambda = \lambda_1^{-1} \Delta_\ell$ with Jacobian λ_1^{p-1} and ${}^1\Delta_\ell = \text{diag}(1, \ell_2, \dots, \ell_p)$. The joint distribution of $\lambda_1, \ell_2, \dots, \ell_p$ becomes:

$$h(\lambda_1, \ell_2, \dots, \ell_p) = C \frac{\pi^{\frac{1}{2}p^2}}{\Gamma_p(\frac{1}{2}p)} \lambda_1^{pn-1} |\Delta_\ell|^{n-\frac{1}{2}(p+1)} |I_{p-1} - \Delta_\ell| \cdot \alpha_{p-1}(\Delta_\ell) \sum_{k=0}^{\infty} \sum_K \frac{(\alpha)_K (\beta)_K}{(\theta)_K} \frac{C_K(-A)}{C_K(I_p)k!} \lambda_1^k C_K({}^1\Delta_\ell),$$

where $\Delta_\ell = \text{diag}(\ell_2, \dots, \ell_p)$ and $\alpha_{p-1}(\Delta_\ell) = \prod_{1 < i < j \leq p} (\ell_i - \ell_j)$.

Therefore $f(\lambda_1) = \int_{\ell_2 > 0} \dots \int_{\ell_p > 0} h(\lambda_1, \ell_2, \dots, \ell_p) d\ell_2, \dots, d\ell_p$.

Hence the result by Sugiyama (1967).

The distribution of the smallest root of matrix $X: p \times p$ follows similarly.

5. The distribution of the ratio of two independent random determinants

Let $X_i: p \times p$, $i=1, 2$ be random matrices independently distributed with the following density function

$$(5.1) \quad f(X_i) = \begin{cases} C_i |X_i|^{n_i - \frac{1}{2}(p+1)} {}_2F_1(\alpha_i, \beta_i; \theta_i; -A_i X_i), & X_i > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{where } C_i = \frac{|A_i|^{n_i} \Gamma_p(\alpha_i) \Gamma_p(\beta_i) \Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\theta_i) \Gamma_p(\alpha_i - n_i) \Gamma_p(\beta_i - n_i)}, \quad i=1, 2.$$

Theorem 5.1

Let X_1 and X_2 be two independent random matrices with density function as given in (5.1) for $i=1, 2$ respectively. Then $v = |X_1| |X_2|^{-1}$ has the density

$$f(v) = \prod_{i=1}^2 \left\{ \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i) \Gamma_p(\beta_i - n_i)} \right\} \pi^{p(p-1)} v^{-1} \\ \cdot G_{4p}^{3p} \left[\begin{matrix} |A_1| \\ |A_2| \end{matrix} \middle| v \begin{matrix} a_1, \dots, a_{4p} \\ b_1, \dots, b_{4p} \end{matrix} \right]$$

where

$$G_p^{r,s} \left[x \middle| \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_p \end{matrix} \right] = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} \frac{\prod_{j=1}^r \Gamma(b_j + h) \prod_{j=1}^s \Gamma(1 - a_j - h)}{q \prod_{j=r+1}^p \Gamma(1 - b_j - h) p \prod_{j=s+1}^p \Gamma(a_j + h)} x^{-h} dh \\ (\text{Mathai (1971)}).$$

Proof:

Consider the h -th moment of v , namely

$$(5.2) \quad E(v^h) = \int_{X_1 > 0} |X_1|^h f(X_1) dX_1 \cdot \int_{X_2 > 0} |X_2|^{-h} f(X_2) dX_2.$$

By using the fact that $\int_{X_i > 0} f(X_i) dX_i = 1$ it follows from (5.1) and (5.2) that

$$(5.3) \quad E(v^h) = \prod_{i=1}^2 \left\{ \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i) \Gamma_p(\beta_i - n_i)} \right\} |A_1|^{-h} |A_2|^{+h} \\ \cdot \left[\frac{\Gamma_p(n_1 + h) \Gamma_p(\alpha_2 - n_2 + h) \Gamma_p(\beta_2 - n_2 + h) \Gamma_p(n_2 - h) \Gamma_p(\alpha_1 - n_1 - h) \Gamma_p(\beta_1 - n_1 - h)}{\Gamma_p(\theta_1 - n_1 - h) \Gamma_p(\theta_2 - n_2 + h)} \right]$$

where the real parts in the $\Gamma(\cdot)$ are positive.

After comparing the multivariate gamma functions in the square brackets in (5.3) it follows that

$$E(v^h) = \prod_{i=1}^2 \left\{ \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i) \Gamma_p(\beta_i - n_i)} \right\} |A_1|^{-h} |A_2|^{+h} \pi^{p(p-1)} \\ \frac{\prod_{j=1}^{3p} \Gamma(b_j + h) \prod_{j=1}^{3p} \Gamma(1 - a_j - h)}{\prod_{j=3p+1}^{4p} \Gamma(1 - b_j - h) \prod_{j=3p+1}^{4p} \Gamma(a_j + h)}$$

where

$$b_j = \begin{cases} n_1 - \frac{1}{6}(j-1); j=1,4,\dots,3p-2. \\ \alpha_2 - n_2 - \frac{1}{6}(j-2); j=2,5,\dots,3p-1 \\ \beta_2 - n_2 - \frac{1}{6}(j-3); j=3,6,\dots,3p \\ n_1 - \theta_1 + \frac{1}{2}(j-3p+1); j=3p+1,\dots,4p \end{cases}$$

and

$$a_j = \begin{cases} 1 - n_2 - \frac{1}{6}(j-1); j=1,4,\dots,3p-2 \\ 1 - \alpha_1 + n_1 - \frac{1}{6}(j-2); j=2,5,\dots,3p-1 \\ 1 - \beta_1 + n_1 - \frac{1}{6}(j-3); j=3,6,\dots,3p \\ \theta_2 - n_2 - \frac{1}{2}(j-3p-1); j=3p+1,\dots,4p \end{cases}$$

Now, by considering the inverse Mellin transform (Erdelyi et. al. (1964)),

$$f(v) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} E(v^h) v^{-h-1} dh$$

we obtain the density function of the random variable v .

6. Distributions of a random matrix with random matrix parameter

We consider the cases where the conditional distribution, assumed to be of a multivariate type defined in (2.5) and (2.6), contains a matrix parameter with an a priori Wishart distribution (Roux (1971)).

Theorem 6.1

Let the random matrix $X: p \times p$ have a conditional distribution with density function

$$(6.1) \quad f(X|P_I) = \frac{|P_I|^n \Gamma_p(\theta-n)}{\Gamma_p(n) \Gamma_p(\theta)} |X|^{n-\frac{1}{2}(p+1)} {}_0F_1(\theta; -P_I X), \quad X > 0.$$

Let the marginal distribution of $P_I: p \times p$ be a Wishart distribution with density function

$$(6.2) \quad g_1(P_1) = \frac{|A|^{n-\beta}}{\Gamma_p(\beta-n)} e^{-\text{tr } A^{-1} P_1} |P_1|^{\beta-n-\frac{1}{2}(p+1)}.$$

Then the unconditional distribution of X is given by the density function (2.5).

Proof:

The unconditional distribution is given by,

$$f_2(X) = \int_{P_1 > 0} f(X|P_1) g_1(P_1) dP_1.$$

The Theorem follows by using (6.1), (6.2) and the integral (Constantine (1963))

$$\int_{S > 0} e^{-\text{tr } RS} |S|^{t-\frac{1}{2}(p+1)} C_K(SW) dS = \Gamma_p(t, K) |R|^{-t} C_K(WR^{-1}).$$

Theorem 6.2

Let the random matrix $X: p \times p$ have a conditional distribution with density function

$$f(X|P_2) = \frac{|P_2|^n \Gamma_p(\beta) \Gamma_p(\theta-n)}{\Gamma_p(n) \Gamma_p(\theta) \Gamma_p(\beta-n)} |X|^{n-\frac{1}{2}(p+1)} {}_1F_1(\beta; \theta; -P_2 X), \quad X > 0$$

and the marginal distribution of $P_2: p \times p$ be a Wishart distribution with density function

$$g_1(P_2) = \frac{|A|^{n-\alpha}}{\Gamma_p(\alpha-n)} e^{-\text{tr } A^{-1} P_2} |P_2|^{\alpha-n-\frac{1}{2}(p+1)}$$

The unconditional distribution of X is given by the density function (2.1).

The proof follows as in theorem 6.1.

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